

LYRICA

Mathematical Background and Processing Pipeline

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4 Overview

LYRICA is a deterministic information-theoretic framework for computational analysis of song lyrics.

The framework does not use machine learning, neural networks, large language models, word embeddings or transformer models. Instead, all descriptors are computed directly from lexical statistics and deterministic rule-based dictionaries.

5 Processing Pipeline

The complete processing workflow consists of the following stages.

1. Reading UTF-8 plain text files.
2. Automatic parsing of albums and songs.
3. Removal of metadata and non-lyrical content.
4. Detection of lyrical blocks and tokenization.
5. Dictionary-based lexical analysis.
6. Statistical analysis.
7. Information-theoretic analysis.
8. Album-level aggregation.
9. Visualization and CSV export.

6 Dictionary-Based Analysis

Each token is analysed using deterministic lexical dictionaries.

Current dictionaries include

- lexical vocabulary,
- action verbs,
- auxiliary verbs,
- modifiers,
- pronouns,
- negation words,
- temporal markers,
- structural connectors.

The dictionaries are evaluated using rule-based processing only. No statistical language model or neural network is employed.

7 Basic Statistical Descriptors

For every song the software computes

- total number of words,
- unique vocabulary,
- lexical density,
- repeated lines,
- repetition ratio,

- Zipf rarity,
- rare-word density,
- structural agency descriptors.

8 Shannon Entropy

Let

$$N$$

be the total number of words in the song and

$$n_i$$

the number of occurrences of the i -th unique word.
The probability of every word is

$$p_i = \frac{n_i}{N}.$$

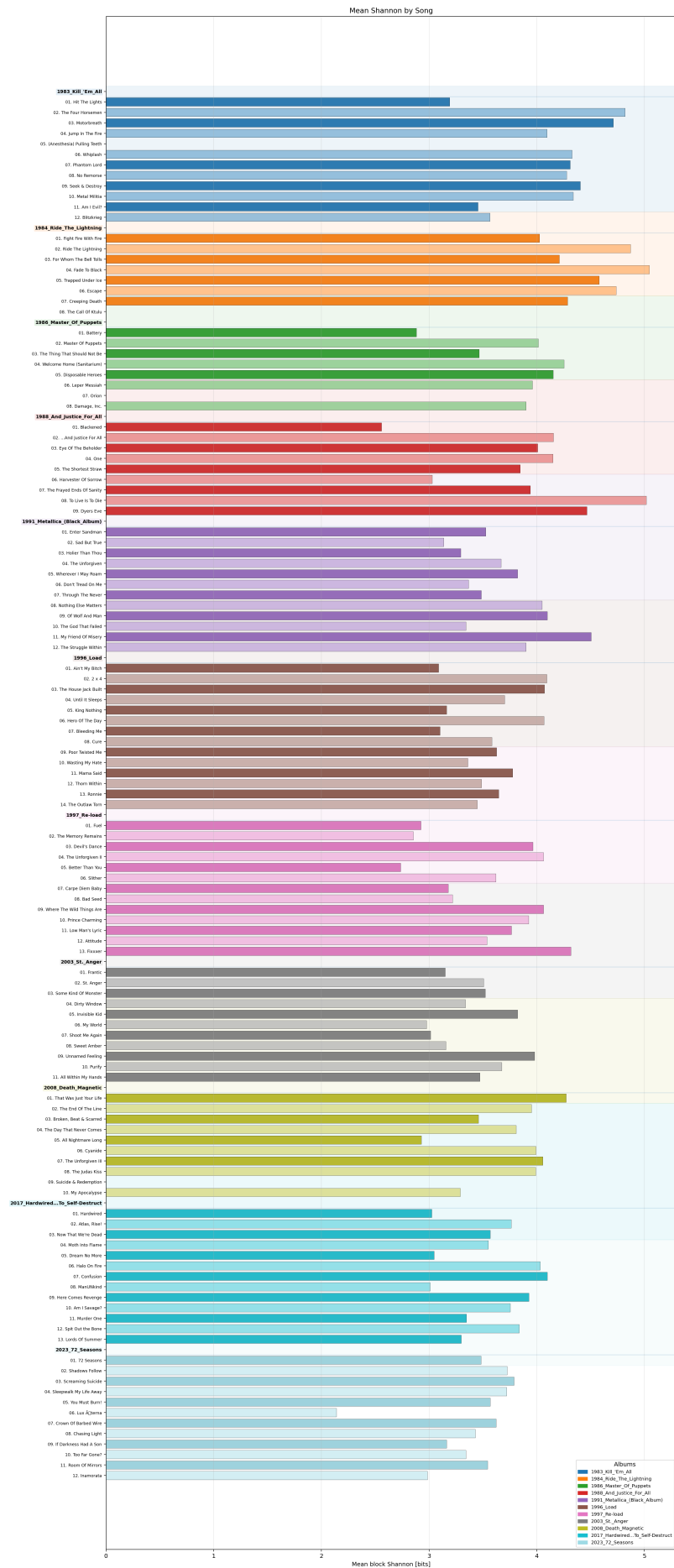
The Shannon entropy is computed as

$$H = - \sum_{i=1}^M p_i \log_2(p_i),$$

where

- M is the number of unique words;
- p_i is the relative occurrence probability.

Higher entropy indicates larger lexical diversity, whereas lower entropy corresponds to stronger repetition.



9 Lexical Fisher Information

The lyrics are divided into consecutive structural blocks.

For every block the Shannon entropy is computed

$$H_1, H_2, \dots, H_K.$$

The discrete Fisher information is

$$F_L = \sum_{i=1}^{K-1} \frac{(H_{i+1} - H_i)^2}{H_i + \varepsilon}.$$

Large values correspond to abrupt structural changes between neighbouring lyrical blocks.

10 Action Fisher

For each lyrical block, the software estimates an *Action Density* descriptor using a deterministic dictionary of action-related lexical units (mainly verbs and action expressions).

Let

$$N_i$$

denote the total number of words in lyrical block i , and

$$N_{A,i}$$

the number of words identified as action-related.

The normalized Action Density is defined as

$$A_i = \frac{N_{A,i}}{N_i}.$$

The sequence

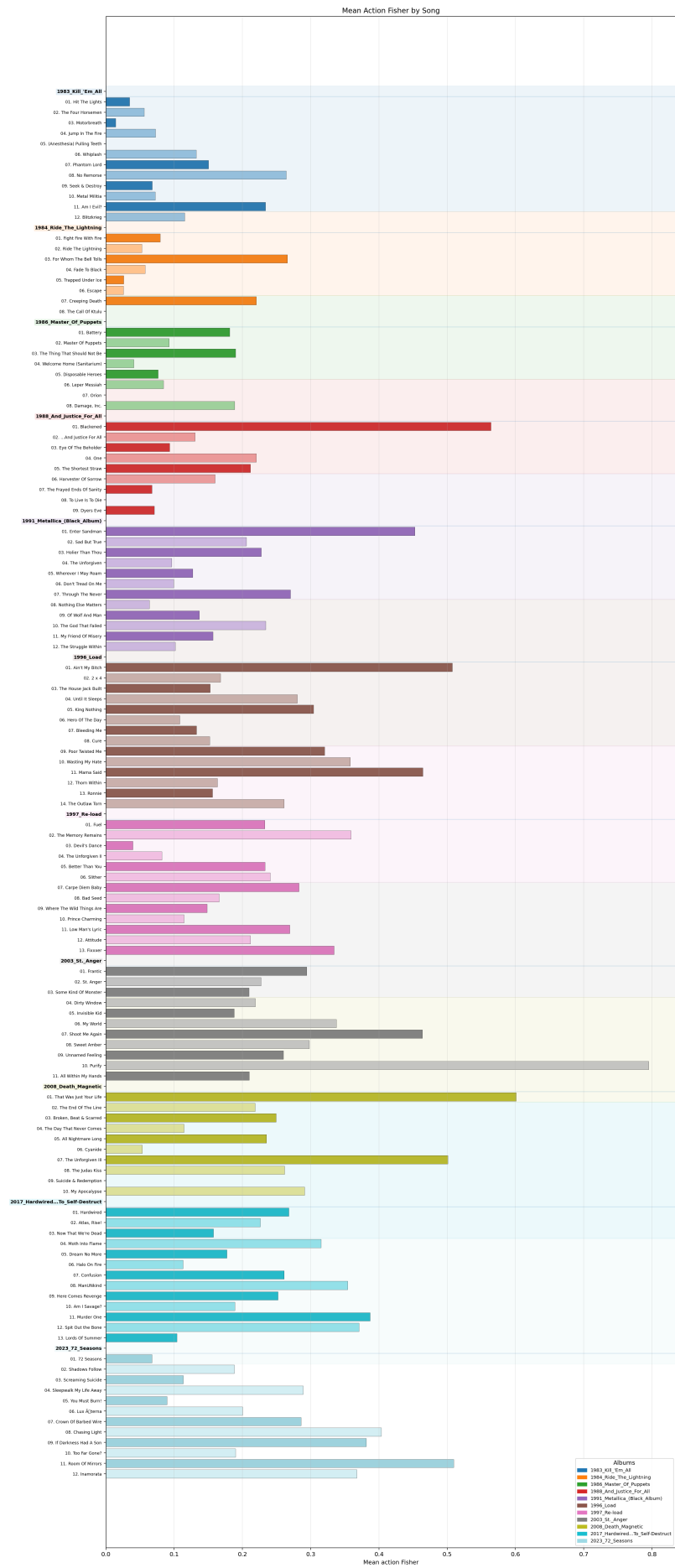
$$A_1, A_2, \dots, A_K$$

describes the temporal evolution of action intensity throughout the song.

The corresponding Fisher descriptor is computed as

$$F_A = \sum_{i=1}^{K-1} \frac{(A_{i+1} - A_i)^2}{A_i + \varepsilon}.$$

Large values of F_A indicate abrupt changes in the distribution of action-related language between neighbouring lyrical blocks, whereas lower values correspond to smoother narrative progression.



11 Gini Coefficient

Album statistics are obtained from all songs belonging to the same album.

After sorting the metric values

$$x_1 \leq x_2 \leq \cdots \leq x_n,$$

the Gini coefficient is

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n \sum_{i=1}^n x_i}.$$

The coefficient measures the inequality of descriptor distribution among songs.

12 Manhattan Distance

For two songs represented by feature vectors

$$\mathbf{x}, \mathbf{y} \in \mathbb{R}^d,$$

the Manhattan distance is

$$D(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^d |x_k - y_k|.$$

Album-level structural dispersion is obtained from all pairwise song distances.

13 Output

The software generates

- song-level descriptors,
- album-level statistics,
- Shannon and Fisher metrics,
- lexical statistics,
- Gini statistics,
- CSV files,
- PNG visualizations.

14 Remarks

The generated descriptors constitute an information-theoretic representation of lyrical structure.

The extracted numerical descriptors are intended exclusively for scientific analysis and statistical comparison of lyrics. They are not sufficient to reconstruct or reproduce the original copyrighted literary work.